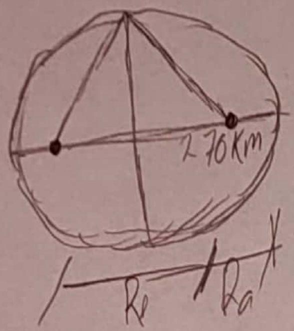
Exámen Corto

1) 360Km * punto más largo
180Km * punto más cercano

a)



$R_p = 180Km$
$R_a = 360Km$

$a^2 = b^2 + c^2$
$c^2 = a^2 - b^2$

$R_m = \frac{R_p + R_a}{2} = a$

$a = \frac{180 + 360}{2} = \underline{270}$

b) $e = c/a$

$e = \frac{a - x}{a} = \frac{270 - 180}{270}$

$e = \frac{1}{3} \approx \underline{0.333}$

2)

$m_1 = 25kg$

$m_2 = 100kg$

$r = 0.2m$

$40m = ad$

$20m = x$

$$fg = \frac{G m_1 m_2}{r^2}$$

$$fg = \frac{(6.67 \times 10^{-11} Nm^2/kg.)(25kg)(100kg)}{(20m)^2}$$

$$fg = \underline{4.168 \times 10^{-10}}$$

3)

$m = 0.2kg$

$q_h = 0.7 l/m$

$D = 1.5$

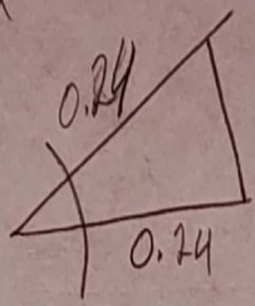
$\Delta = 0.2\cancel{4}$

$T = 0$

$d = x$

$X = A\cancel{\cos}(wt + \theta)$

$f - kx$



$T = \frac{1}{f}$

$f = \frac{1}{T}$

$f = \frac{1}{1.5 seg}$

$f = \frac{2}{3}$